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Department of Information and Communication Technology

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Project Report
Web Technologies Laboratory (ICE-3206)

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Date of Submission: 09 November 2026

APPROVAL

This is to certify that the project entitled **"Smart Hall Meal Management System"** has been carried out by the students listed below under my supervision as a part of the academic requirements of **Web Technologies Laboratory (ICE-3206)** at **Information and Communication Technology, Bangladesh University of Professionals (BUP) .**

The project focuses on developing a web-based platform to organize and simplify the daily meal management activities of a university residential hall. The system includes features for student management, meal booking, meal verification, bazar expense recording, meal rate calculation, balance management, and administrative monitoring.

To the best of my knowledge, the project has been completed by the students as part of their academic work. It is hereby approved for submission and evaluation.

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DECLARATION

We hereby declare that the project entitled **"Smart Hall Meal Management System"** is an original work carried out by us as part of the academic requirements of **Web Technologies Laboratory (ICE-3206)** at **Information and Communication Technology, Bangladesh University of Professionals (BUP) .**

We confirm that the project has been developed through our own efforts under the guidance and supervision of Md. Istakiak Adnan Palash, Lecturer, Dept. of ICT, BUP and Sharmeen Jahan Seema, Lecturer, Dept. of ICT, BUP .

The system has been designed and implemented to address practical challenges associated with daily meal management in a university residential hall.

We further declare that this project has not been submitted previously, either in whole or in part, to any other institution for the award of any academic degree or qualification. All external sources and materials used in preparing this report have been properly acknowledged in the References section.

We accept responsibility for the authenticity and accuracy of the information presented in this report.

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ACKNOWLEDGEMENT

First and foremost, we would like to express our sincere gratitude to Almighty Allah for giving us the strength, patience, and determination to complete this project successfully.

We would like to express our heartfelt appreciation to our project supervisor, Md. Istakiak Adnan Palash, Lecturer, Dept. of ICT, BUP and Sharmeen Jahan Seema, Lecturer, Dept. of ICT, BUP for the valuable guidance, constructive suggestions, and continuous support throughout the development of the Smart Hall Meal Management System. Their feedback and recommendations helped us improve the system and overcome various technical challenges during the development process.

We would also like to thank everyone who participated in testing the application and provided feedback on its usability and functionality. Their cooperation helped us identify issues and improve different parts of the system.

Finally, we would like to acknowledge the contributions of team members. Their combined efforts, cooperation, and dedication made it possible to complete the project and prepare this report.

PREFACE

This project report has been prepared as part of the academic work on the **Smart Hall Meal Management System**. The project focuses on developing a web-based application to improve the management of daily meal-related activities in a university residential hall.

The development of this system provided an opportunity to apply theoretical knowledge of software engineering and web development to a practical problem. Throughout the project, we worked on different aspects of software development, including requirement analysis, system design, frontend and backend development, database management, deployment, and testing.

The report presents the overall development process of the system, beginning with the identification of existing meal management problems and followed by the analysis, design, implementation, and testing of the proposed solution. It also describes the technologies used and the major features developed to support students, mess staff, and administrators.

We have made an effort to present the project activities, technical decisions, and implementation details in a clear and organized manner. We hope this report provides a proper understanding of the system, its functionality, and its potential contribution to making residential hall meal management more organized and efficient.

ABSTRACT

Managing daily meals in a university residential hall involves several activities, including meal booking, maintaining student records, calculating meal expenses, managing balances, and verifying meal distribution. When these activities are handled manually, the process can become time-consuming and difficult to maintain accurately. Errors in record keeping, delays in updating information, and difficulties in calculating individual meal costs can create problems for both students and hall management.

The **Smart Hall Meal Management System** is a web-based application developed to make these activities more organized and efficient. The system provides separate access for administrators, mess staff, and students, with permissions based on their respective roles. Students can register for an account, manage their profiles, select meals before the designated cutoff time, and submit requests for balance deposits. Administrators can approve student accounts, manage users, control meal cutoff times, review deposit requests, and monitor the overall activities of the system.

The system also includes a meal verification process that allows mess staff to identify students and mark meals as served. Daily bazar expenses can be recorded by the staff, and the system calculates the meal rate based on the recorded expense and the number of students who selected the particular meal. Any remaining amount caused by division is preserved and included in the report. Staff and administrators can also download relevant reports in CSV format for further use.

The application was developed using React.js for the frontend, Node.js and Express.js for the backend, and MongoDB for database management. The system follows a three-tier architecture to separate the user interface, application logic, and data storage.

Functional and validation testing was performed on the major modules of the application. The testing process covered account management, meal booking, meal verification, expense calculation, balance management, and other essential operations. The developed system aims to reduce manual work, improve record accuracy, and provide a more transparent and manageable approach to residential hall meal administration.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
CSS	Cascading Style Sheets
DB	Database
DBMS	Database Management System
ERD	Entity Relationship Diagram
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
JS	JavaScript
MERN	MongoDB, Express.js, React.js, Node.js
QR	Quick Response
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

CHAPTER 1: INTRODUCTION

1.1 Background of the Project

Managing daily meals in a university residential hall requires coordination among students, mess staff, and hall administrators. Every day, students need to decide whether they will take breakfast, lunch, or dinner. At the same time, the mess staff must prepare meals according to the number of students who have selected each meal and maintain records of daily market expenses.

In many traditional hall management systems, these activities are performed using paper-based records, spreadsheets, or informal communication. Although these methods may work for a small number of students, they become difficult to manage when the number of residents increases. Maintaining accurate meal records, calculating individual expenses, tracking student balances, and verifying meal distribution can require considerable time and effort.

The Smart Hall Meal Management System was developed to address these challenges through a centralized web-based platform. It provides a common system where students, mess staff, and administrators can perform their respective activities. Each user receives access to the features relevant to their role.

Students can select their meals within the permitted time, check their account information, and request balance deposits. Mess staff can record daily bazar expenses and verify meal distribution. Administrators can manage user accounts, approve registration requests, control meal cutoff times, and monitor different activities.

The system also automates the calculation of meal rates based on the recorded bazar expenses and the number of students who selected a particular meal. This reduces the need for manual calculations and helps maintain more consistent financial records.

Overall, the project aims to provide a practical and organized solution for managing the daily meal-related activities of a university residential hall.

1.2 Problem Statement

Traditional meal management processes often depend on manual record keeping and communication between students and hall staff. These processes can create several difficulties in daily operations.

One of the major problems is maintaining accurate meal selection records. When students communicate their meal preferences manually, updating and counting the total number of meals can become difficult. This may result in incorrect meal preparation estimates or inconsistencies in the records.

Another important challenge is the calculation of individual meal costs. Daily bazar expenses need to be distributed among the students who have selected the corresponding meal. Performing these calculations manually can lead to errors, particularly when the number of students and daily expenses change frequently.

Managing student balances and deposit requests is also difficult when financial information is maintained separately. Similarly, verifying whether a student has already received a meal can be challenging without a centralized record.

Therefore, a system is needed that can organize meal selection, automate meal rate calculation, maintain student balances, and support meal verification through a single platform. The Smart Hall Meal Management System has been developed to address these issues.

1.3 Objectives of the Project

The main objective of this project is to develop a web-based platform that simplifies and organizes the meal management activities of a university residential hall.

The specific objectives are:

1. To develop a centralized platform for managing student meal-related activities.
2. To provide separate dashboards and access permissions for administrators, mess staff, and students.
3. To allow students to register and access the system after administrative approval.
4. To enable students to select or cancel meals before the designated cutoff time.
5. To allow administrators to manage meal cutoff times according to operational requirements.
6. To implement a meal verification process to maintain records of served meals and prevent duplicate meal distribution.
7. To provide a facility for recording daily bazar expenses.

8. To automate meal rate calculation based on daily expenses and the number of students selecting each meal.
9. To maintain student balances and manage deposit requests through an approval process.
10. To provide facilities for profile management, complaint submission, and notice publication.
11. To generate downloadable reports in CSV format for administrative and financial purposes.
12. To reduce manual workload and improve the accuracy and transparency of hall meal management.

1.4 Scope of the Project

The Smart Hall Meal Management System focuses on the major activities associated with daily meal management in a university residential hall. The system is designed to support three categories of users: administrators, mess staff, and students.

The scope of the project includes the following areas:

Student Management: Students can request new accounts by providing the required personal information and identification details. Administrators can review and approve registration requests. Approved users can access the system and manage their permitted profile information.

Meal Booking: Students can select breakfast, lunch, and dinner according to their requirements. Meal selection is controlled by predefined cutoff times, which can be managed by administrators.

Meal Verification: Mess staff can identify a student using the available identification or verification process and mark a selected meal as served. The system maintains the meal status to prevent the same meal from being recorded as served more than once on the same day.

Bazar and Meal Rate Management: Mess staff can record daily bazar expenses. The system calculates the rate of each meal by dividing the recorded expense by the number of students who selected that meal. Any remaining amount caused by division is preserved in the report.

Balance Management: Students can submit requests to add money to their accounts. The amount is added to the student's balance only after administrative approval. The system also applies a balance restriction when a student's negative balance exceeds the configured limit.

Complaint and Notice Management: Students can submit complaints through the platform, while administrators can review and respond to them. Administrators can also publish notices for system users.

Report Generation: Staff and administrators can download relevant meal and expense reports in CSV format for record keeping and further analysis.

The project is primarily intended for web-based use. It does not include an integrated online payment gateway in the current version.

1.5 Motivation

The motivation behind developing this project comes from the practical difficulties associated with managing daily meals in a residential hall. Meal selection, expense calculation, balance management, and meal distribution are closely connected activities. When these activities are handled separately, maintaining accurate and updated information becomes difficult and even some days few students had taken meal of other's student without their acknowledgement .

A centralized digital system can reduce unnecessary manual work and make information more accessible to the people responsible for managing hall meals. It can also help students understand their meal records and financial status more clearly.

Another motivation was to apply the knowledge gained from academic courses to a practical software development project. Through this project, we aimed to gain hands-on experience in frontend development, backend programming, database management, authentication, system deployment, and software testing.

The project therefore serves two purposes: solving a practical management problem and providing an opportunity to develop technical skills through the implementation of a real-world application.

1.6 Organization of the Report

This report is organized into six chapters to present the development process and overall functionality of the Smart Hall Meal Management System.

Chapter 1: Introduction provides the background, problem statement, objectives, scope, and motivation behind the project.

Chapter 2: Literature Review and Existing System discusses the traditional meal management process, its limitations, and the proposed solution.

Chapter 3: System Analysis and Design presents the system requirements, user roles, architecture, use case diagram, data flow diagram, and database design.

Chapter 4: System Implementation explains the technologies, development process, and implementation of the major modules of the application.

Chapter 5: Testing and Results describes the testing methods used to evaluate the system and discusses the results obtained from the testing process.

Chapter 6: Conclusion and Future Work summarizes the project, highlights its achievements and limitations, and identifies possible improvements for future development.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SYSTEM

2.1 Introduction

Meal management is an important administrative activity in university residential halls. It involves several interconnected tasks, such as collecting meal preferences, estimating food requirements, recording daily expenses, calculating meal costs, and maintaining student financial records. Proper coordination of these activities is necessary to ensure that meals are distributed efficiently and expenses are managed accurately.

Traditionally, many residential hall meal management activities have been handled through manual records, paper-based registers, spreadsheets, and direct communication between students and mess staff. Although these methods can support basic operations, they often require repeated manual work and make it difficult to maintain updated information.

With the development of web-based technologies and database management systems, these activities can be organized through centralized digital platforms. Such systems can provide structured data storage, controlled user access, automated calculations, and easier access to relevant information.

The Smart Hall Meal Management System follows this approach by integrating the major meal-related activities into a single web-based application. This chapter discusses the traditional management process, its limitations, and the proposed solution.

2.2 Overview of Traditional Hall Meal Management

In a traditional residential hall environment, meal management generally involves students, mess staff, and administrative personnel. Students communicate their meal preferences, while mess staff estimate the required food quantity and manage daily meal distribution.

Meal selections are often collected through paper records, messaging groups, or other informal methods. Staff members then count the selected meals and prepare food according to the available information.

Daily market expenses are usually recorded separately. At the end of a particular period, the total expense is distributed among the students based on their meal participation. This requires maintaining accurate records of both expenses and meal counts.

Student deposits and balances may also be maintained through separate registers or spreadsheets. As a result, updating financial information and checking individual balances can require additional effort.

Although this process can be followed in smaller environments, maintaining consistency becomes increasingly difficult when multiple activities need to be managed simultaneously.

2.3 Limitations of the Existing System

Traditional meal management methods have several limitations that can affect the efficiency and accuracy of daily operations.

1. Manual Meal Recording

Collecting and maintaining meal selections manually requires continuous attention from students and staff. Changes in meal preferences may not always be reflected correctly in the records.

2. Difficulty in Meal Counting

Staff members need to count the number of students selecting each meal. When records are maintained separately, calculating the total number of meals can become time-consuming.

3. Manual Expense Calculation

Daily bazar expenses need to be distributed among the students who selected the corresponding meals. Manual calculations may introduce errors and make it difficult to maintain consistent financial records.

4. Lack of Centralized Information

Student details, meal selections, expenses, and financial records may be stored in different places. This makes it difficult for authorized users to access all relevant information from one location.

5. Meal Distribution Verification

Without a structured verification process, it may be difficult to confirm whether a student has already received a particular meal. This can create inconsistencies in meal distribution records.

6. Difficulty in Balance Management

Maintaining individual student balances and processing deposit requests manually requires additional administrative effort. It may also create delays when students need updated financial information.

7. Limited Reporting Facilities

Preparing daily or monthly reports from separate records can be time-consuming. It may also require additional calculations before the information can be used for administrative purposes.

These limitations indicate the need for a centralized system that can organize meal-related information and reduce unnecessary manual activities.

2.4 Proposed System

The Smart Hall Meal Management System is proposed as a web-based solution for managing the major activities of a university residential hall. It provides a centralized platform where students, mess staff, and administrators can interact with the system according to their assigned roles.

The proposed system organizes meal booking, student management, meal verification, bazar expense recording, meal rate calculation, balance management, complaint handling, notice publication, and report generation.

The major components of the proposed system are described below.

1. Student Management

Students can register by submitting the required information. Administrators can review registration requests and approve accounts before students receive access to the system. Approved students can manage their permitted profile information.

2. Meal Booking Management

Students can select their required meals through the application. Meal selection is controlled by predefined cutoff times to prevent late changes after the designated period.

3. Meal Verification

The system provides a verification process for recording meal distribution. Mess staff can identify students and update the status of a selected meal as served. This helps maintain a record of meal distribution and reduces the possibility of duplicate serving entries.

4. Bazar Management

Mess staff can record daily bazar expenses through the application. These records are used to calculate the cost of meals for the corresponding period.

5. Automated Meal Rate Calculation

The system calculates the meal rate by dividing the recorded expense by the number of students who selected the corresponding meal. This reduces the need for repeated manual calculations and helps maintain consistent records.

6. Balance and Deposit Management

Students can submit deposit requests through the system. Administrators can review these requests and approve or reject them. Approved deposits are reflected in the respective student accounts.

7. Complaint and Notice Management

Students can submit complaints through the application, and administrators can review and respond to them. Administrators can also publish notices to communicate important information to users.

8. Report Generation

The system provides report generation facilities, including CSV downloads, allowing authorized users to maintain records and review relevant information.

2.5 Comparison Between Existing and Proposed Systems

The following table presents a comparison between traditional meal management methods and the proposed Smart Hall Meal Management System.

Table 2.1: Comparison Between Existing and Proposed Systems

Criteria	Existing System	Proposed System
Meal Booking	Manual or informal communication	Web-based meal selection
Meal Records	Paper or separate records	Centralized database
Meal Counting	Manual counting	System-based meal records
Cutoff Management	Difficult to control consistently	Configurable cutoff time
Meal Verification	Manual checking	Digital meal verification
Expense Recording	Separate registers or spreadsheets	Centralized bazar records
Meal Rate Calculation	Manual calculation	Automated calculation
Balance Management	Separate financial records	Digital student balance records
Deposit Requests	Manual communication	Online request and approval
Complaint Management	Direct communication	Centralized complaint module
Notice Publication	Physical notice or messaging	Digital notice management
Report Generation	Manual preparation	Downloadable CSV reports

The comparison shows that the proposed system provides a more centralized and organized approach to meal management. It reduces dependence on separate manual records and provides specific facilities for handling different administrative activities.

However, the proposed system is intended to support existing hall management operations rather than replace all human involvement. Administrative approval, expense entry, and meal distribution still require authorized users to perform their respective responsibilities.

2.6 Summary

This chapter discussed the traditional approach to residential hall meal management and identified several limitations associated with manual record keeping, meal counting, expense calculation, balance management, and report preparation.

The proposed Smart Hall Meal Management System addresses these challenges through a centralized web-based platform. By integrating student management, meal booking, verification, bazar recording, financial management, and reporting facilities, the system aims to make daily meal administration more organized and manageable.

The next chapter presents the system requirements and design, including user roles, system architecture, diagrams, and database structure.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

System analysis and design is an essential phase in software development. It involves identifying user requirements, defining system functionalities, designing the system architecture, and determining how different components interact with one another.

The Smart Hall Meal Management System was designed to provide a centralized platform for managing hall meal-related activities. The system aims to reduce manual work, improve record management, simplify meal booking, and make meal distribution more organized.

The system supports three primary user roles: Admin, Staff, and Student. Each role has specific responsibilities and access permissions. The application follows a three-tier architecture consisting of the presentation layer, application layer, and database layer.

This chapter presents the requirement analysis, functional and non-functional requirements, user roles, system architecture, use case model, data flow diagram, and database design of the proposed system.

3.2 Requirement Analysis

Requirement analysis was conducted to identify the necessary features and operational expectations of the Smart Hall Meal Management System. The requirements were determined based on the common challenges of traditional hall meal management and the activities required by different categories of users.

The system requirements are classified into two major categories:

1. Functional Requirements: These describe the specific operations and services that the system must provide to its users.

2. Non-Functional Requirements: These describe the quality attributes, performance expectations, security considerations, and operational characteristics of the system.

The requirements are further organized according to the three primary user roles to establish a clear understanding of their respective responsibilities.

3.3 Functional Requirements

Functional requirements define the specific operations and services provided by the Smart Hall Meal Management System. Since the application supports three primary user roles, the functional requirements are categorized according to the responsibilities and permissions of Admin, Staff, and Student.

3.3.1 Admin Functional Requirements

Admin is responsible for managing the overall system, controlling user access, monitoring meal-related activities, and maintaining administrative records.

Table 3.1: Admin Functional Requirements

ID	Functional Requirement	Description
FR-01	Admin Authentication	Allows Admin to securely log in and log out of the system.
FR-02	Student Account Approval	Allows Admin to review, approve, or reject student registration requests.
FR-03	User Management	Allows Admin to view and manage registered user accounts.
FR-04	Role and Permission Management	Allows Admin to manage user access according to assigned roles and permissions.
FR-05	Cutoff Time Management	Allows Admin to configure meal booking and cancellation deadlines.
FR-06	Bazar Entry Management	Allows Admin to record and manage daily bazar expenses.
FR-07	Meal Rate Calculation	Calculates meal rates based on recorded expenses and the number of students who selected the corresponding meal.
FR-08	Deposit Request Management	Allows Admin to review, approve, or reject student deposit requests.
FR-09	Complaint Management	Allows Admin to review student complaints and provide responses.
FR-10	Notice Management	Allows Admin to publish and manage notices for system users.

FR-11	Report Generation	Allows Admin to generate reports containing relevant meal, expense, and financial information.
FR-12	CSV Report Export	Allows Admin to download supported reports in CSV format.

3.3.2 Staff Functional Requirements

Staff members are primarily responsible for verifying student meal eligibility and recording meal-serving activities.

Table 3.2: Staff Functional Requirements

ID	Functional Requirement	Description
FR-13	Staff Authentication	Allows Staff to securely log in and log out of the system.
FR-14	QR Code Scanning	Allows Staff to scan student QR codes during meal distribution.
FR-15	Meal Eligibility Verification	Allows Staff to verify whether a student has selected the corresponding meal.
FR-16	Meal Serving Record	Records successful meal verification and helps prevent duplicate serving entries.
FR-17	Profile Management	Allows Staff to view and update permitted personal profile information.
FR-18	Notice Viewing	Allows Staff to view notices published through the system.

3.3.3 Student Functional Requirements

Students use the system to manage their personal accounts, select meals, monitor their balances, and communicate with the hall administration.

Table 3.3: Student Functional Requirements

ID	Functional Requirement	Description
FR-19	Student Registration	Allows students to create accounts by providing the required information.

FR-20	Student Authentication	Allows approved students to securely log in and log out of the system.
FR-21	Password Reset	Provides a temporary password reset mechanism for students who cannot access their accounts.
FR-22	Profile Management	Allows students to view and update personal information and profile pictures.
FR-23	Password Updating	Allows students to change their existing account passwords.
FR-24	Meal Booking	Allows students to select meals for available meal schedules.
FR-25	Meal Cancellation	Allows students to cancel previously selected meals within the permitted time.
FR-26	Meal Booking Status	Allows students to view their meal selection information and booking status.
FR-27	Balance Viewing	Allows students to view their available account balance.
FR-28	Deposit Request Submission	Allows students to submit requests for adding money to their accounts.
FR-29	Complaint Submission	Allows students to submit complaints regarding hall meal-related issues.
FR-30	Notice Viewing	Allows students to view notices published by Admin.

These role-specific functional requirements define the responsibilities and available operations of each user category. They also establish a structured basis for system implementation and subsequent functional testing.

3.4 Non-Functional Requirements

Non-functional requirements describe the expected quality attributes and operational characteristics of the system. These requirements ensure that the application remains usable, secure, reliable, and maintainable.

Table 3.4: Non-Functional Requirements of the System

ID	Requirement	Description
NFR-01	Usability	The system should provide a simple and understandable user interface for all categories of users.
NFR-02	Security	The system should implement authentication and role-based access control to restrict unauthorized access.
NFR-03	Reliability	The system should maintain consistent records during normal operations.
NFR-04	Performance	The system should process user requests within a reasonable time under normal usage conditions.
NFR-05	Maintainability	The system should have a structured codebase that supports future modifications and improvements.
NFR-06	Data Integrity	The system should maintain accurate and consistent user, meal, and financial records.
NFR-07	Accessibility	The system should be accessible through commonly used modern web browsers.
NFR-08	Scalability	The system architecture should allow future expansion of features and user capacity.
NFR-09	Compatibility	The application should support commonly used desktop and mobile web browsers.
NFR-10	Data Availability	The application should provide access to stored information when the required services are available.

3.5 User Roles and Permissions

The Smart Hall Meal Management System supports three primary user roles: Admin, Staff, and Student. Each role has specific responsibilities and permissions based on its operational requirements.

Table 3.5: User Roles and Responsibilities

User Role	Responsibilities and Permissions
Admin	Manages user accounts, approves student registrations, controls meal cutoff time, manages notices, reviews deposit requests, handles complaints, manages bazar expenses, monitors meal rates, and generates reports.
Staff	Scans student QR codes, verifies meal eligibility, and records meal-serving activities.
Student	Manages personal profiles, books or cancels meals, views meal information and account balance, submits deposit requests, and communicates with Admin through complaints.

The system applies Role-Based Access Control (RBAC) to ensure that users can access only the features and operations permitted for their assigned roles. This separation helps maintain operational control and reduces unauthorized access to administrative functions.

3.6 System Architecture

The Smart Hall Meal Management System follows a three-tier architecture. This architecture separates the user interface, application logic, and data storage into different layers. Each layer performs specific responsibilities and communicates with the other layers to complete system operations.

3.6.1 Presentation Layer

The Presentation Layer represents the frontend of the application. It provides the user interface through which Admin, Staff, and Students interact with the system.

The frontend was developed using React.js. It includes role-specific dashboards, forms, navigation components, and interactive elements. Users can perform their permitted activities through these interfaces.

3.6.2 Application Layer

The Application Layer handles the business logic and communication between the frontend and the database.

The backend was developed using Node.js and Express.js. It processes user requests, manages authentication and authorization, handles meal bookings, processes deposit requests, calculates meal rates, and performs other system operations.

The application uses session-based authentication to manage user login sessions and maintain authenticated access.

3.6.3 Database Layer

The Database Layer stores and manages the information required by the application.

MongoDB is used as the database management system, while MongoDB Atlas provides cloud-based database hosting. The database stores user information, meal-related records, bazar expenses, balance information, deposit requests, complaints, notices, and other necessary application data.

3.6.4 Deployment Architecture

The frontend and backend application are deployed through Render, while the database is hosted on MongoDB Atlas. This deployment arrangement allows users to access the application through a web browser without installing the software locally.

The deployment environment connects the application server with the cloud database to support data storage and retrieval during normal system operations.

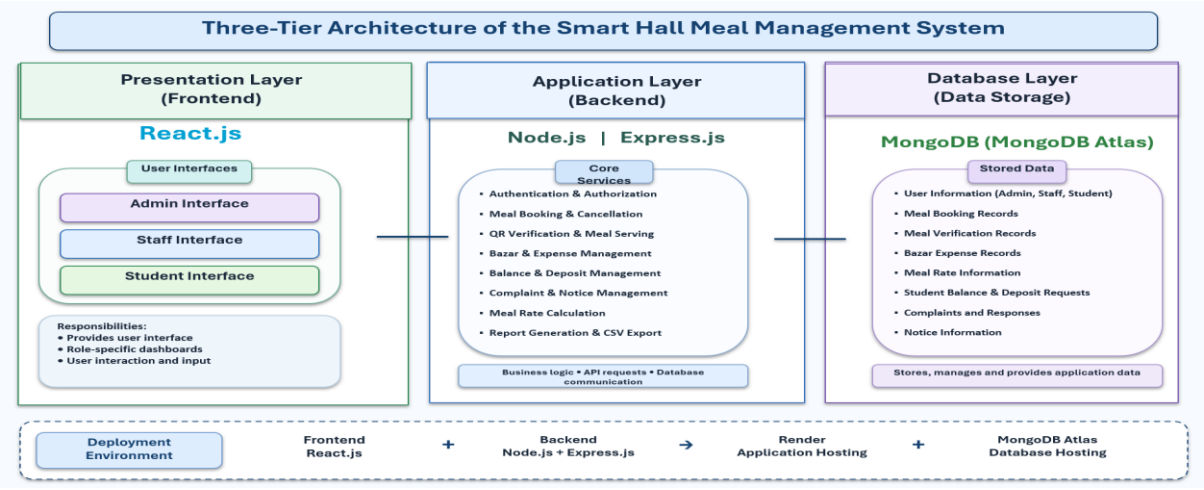


Figure 3.1: Three-Tier Architecture of the Smart Hall Meal Management System

3.7 Use Case Diagram

A Use Case Diagram illustrates the interactions between external actors and the system. It identifies the primary actors and the operations they can perform.

The Smart Hall Meal Management System contains three primary actors:

Admin: Admin manages the overall application, including user accounts, registration approvals, meal settings, bazar expenses, deposit requests, complaints, notices, and reports.

Staff: Staff members perform meal verification activities by scanning student QR codes and recording successful meal-serving operations.

Student: Students interact with the system to register accounts, manage personal information, select meals, view balances, submit deposit requests, and communicate with the administration.

The Use Case Diagram provides a visual representation of the functional scope of the system and the relationships between the actors and their respective operations.

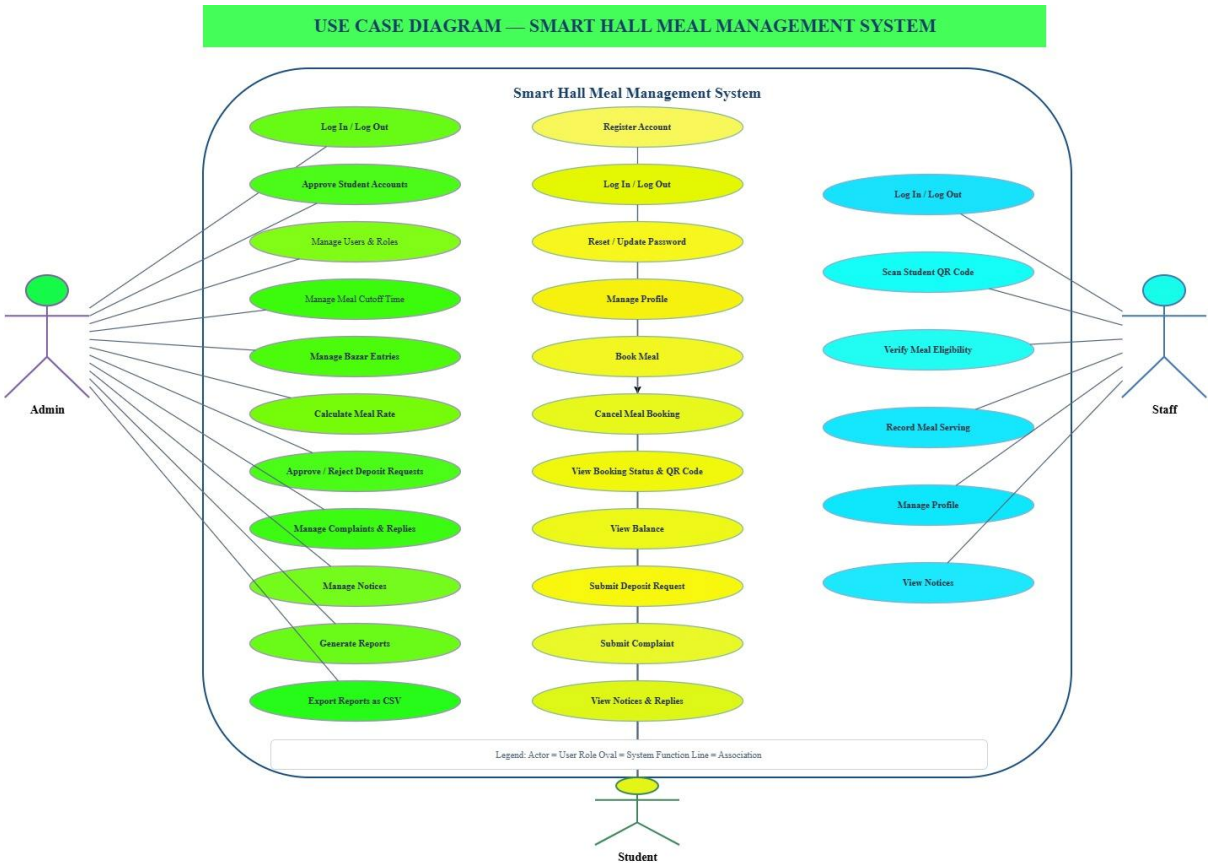


Figure 3.2: Use Case Diagram of the Smart Hall Meal Management System

Link: https://tinyurl.com/4unph4fd

3.8 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) represents how information moves through a system. It illustrates the interaction between external entities, system processes, and data storage.

3.8.1 DFD Level 0 (Context Diagram)

The Level 0 DFD represents the entire Smart Hall Meal Management System as a single process. It shows how the three primary external entities interact with the application.

The major data flows are described below:

Admin: Provides user management information, meal settings, bazar expenses, notices, and administrative decisions. The system returns relevant records, updated information, and reports.

Staff: Sends meal verification information through QR code scanning. The system processes the request and returns the verification result.

Student: Sends registration information, meal selections, deposit requests, and complaints. The system provides account information, meal status, balance information, notices, and relevant responses.

The database maintains the information required to support these operations.

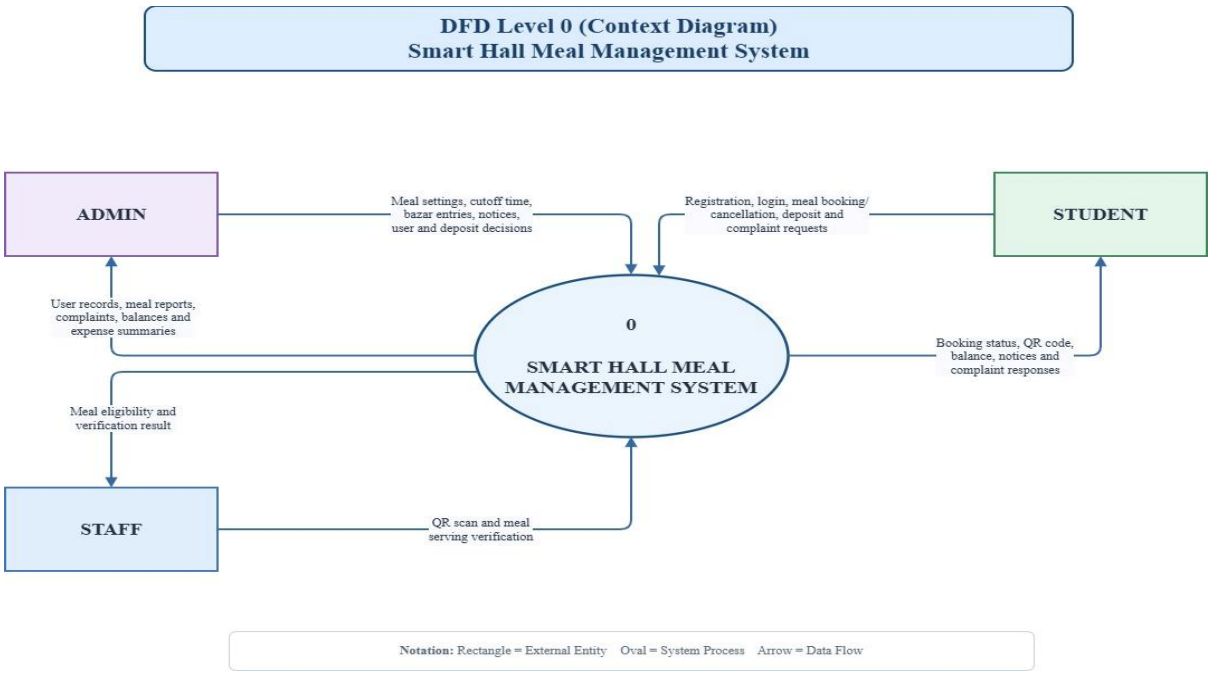


Figure 3.3: Level 0 Data Flow Diagram of the Smart Hall Meal Management System

Link: https://tinyurl.com/yc5h9vt2

3.9 Database Design

Database design is an essential part of the system because it ensures that information can be stored, retrieved, and managed efficiently.

The Smart Hall Meal Management System uses MongoDB, a NoSQL document-oriented database. Unlike traditional relational databases, MongoDB stores information in flexible document structures.

The database is designed to support the following major categories of information:

1. **User Information:** Stores account-related information, user roles, profile details, and authentication-related data.
2. **Meal Booking Information:** Maintains student meal selections and related booking information.
3. **Meal Verification Information:** Stores records associated with QR-based meal verification and serving activities.
4. **Bazar Expense Information:** Maintains daily bazar expense records.
5. **Meal Rate Information:** Supports the calculation and storage of meal rate-related information.
6. **Balance Information:** Stores student balance-related information and financial records.
7. **Deposit Request Information:** Maintains student deposit requests and their approval status.
8. **Complaint Information:** Stores complaints submitted by students and responses from Admin.
9. **Notice Information:** Maintains notices published by Admin.

These data categories collectively support the major operational activities of the application. The document-based database structure provides flexibility for managing different types of application data.

3.10 Summary

This chapter presented the system analysis and design of the Smart Hall Meal Management System. It discussed the requirement analysis, thirty functional requirements categorized according to Admin, Staff, and Student, and ten non-functional requirements.

The chapter also described the user roles and permissions, three-tier system architecture, use case model, data flow diagram, and database design.

The proposed architecture provides a structured foundation for implementing the major functionalities of the application. The next chapter focuses on the technologies, development environment, and practical implementation of the system.

CHAPTER 4: SYSTEM IMPLEMENTATION

4.1 Introduction

System implementation is the phase in which the proposed system design is transformed into a functional software application. It involves selecting appropriate technologies, developing the user interface, implementing application logic, integrating the database, and deploying the application for practical use.

The Smart Hall Meal Management System was developed as a web-based application to simplify and organize hall meal management activities. The application provides separate interfaces for Admin, Staff, and Student users, allowing each category to perform its assigned operations.

This chapter discusses the technologies used, development environment, frontend and backend implementation, database integration, major system modules, and application deployment.

4.2 Technologies Used

Several technologies and development tools were used to develop the Smart Hall Meal Management System. The selected technologies support frontend development, server-side operations, database management, and cloud deployment.

Table 4.1: Technologies Used in the System

Technology	Purpose
React.js	Used to develop the frontend and interactive user interfaces.
JavaScript	Used for implementing client-side and server-side application logic.
Node.js	Provides the runtime environment for backend development.
Express.js	Used to develop backend routes and handle HTTP requests.
MongoDB	Used as the NoSQL database for storing application data.
MongoDB Atlas	Provides cloud-based database hosting and management.
HTML5	Provides the structural elements of the web application.
CSS3	Used for styling and designing the user interface.
Git and GitHub	Used for source code management and version control.
Render	Used for deploying and hosting the web application.

4.3 Development Environment

The development environment consists of the software tools and platforms used during the development and deployment of the application.

The primary development tools include:

- **Visual Studio Code:** Used as the primary code editor for writing and managing the source code.
- **Web Browser:** Used to access the application and perform functional testing.
- **Git:** Used for tracking source code changes.
- **GitHub:** Used to maintain the project repository and manage code updates.
- **MongoDB Atlas:** Used to configure and manage the cloud database.
- **Render:** Used to deploy the application and make it accessible through a public web address.

The development process involved writing the frontend and backend code, integrating the database, testing individual features, and deploying the completed application.

4.4 Frontend Implementation

The frontend represents the user-facing part of the application. It was developed using React.js to provide an interactive and component-based user interface.

The frontend contains separate dashboards and interfaces for Admin, Staff, and Student users. These interfaces are designed according to the responsibilities and permissions of each user role.

The major frontend components include:

4.4.1 Login Interface

The Login Interface allows registered users to access the application by providing their login credentials. After successful authentication, users are directed to the appropriate dashboard according to their assigned roles.

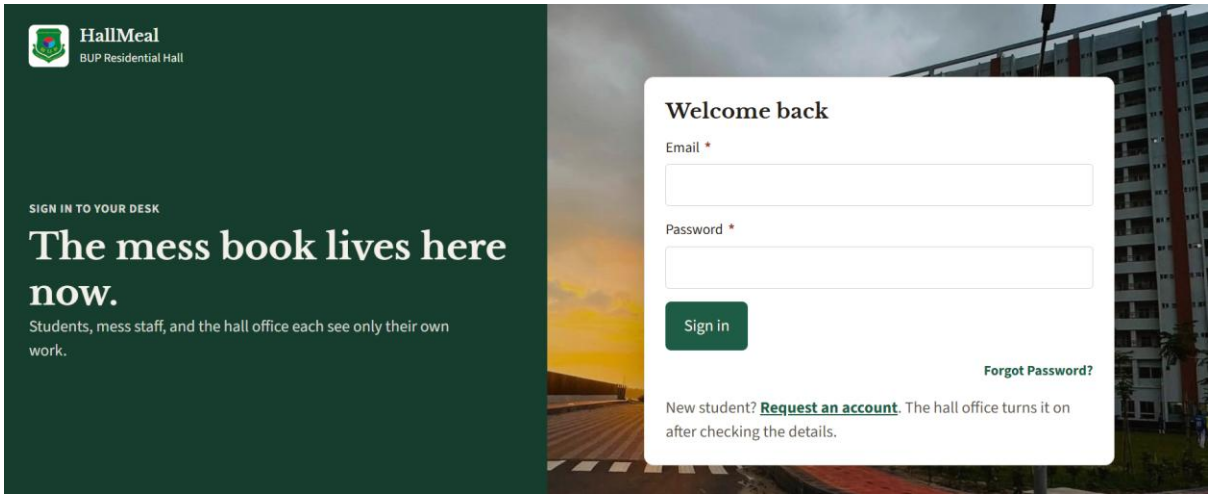


Figure 4.1: Login Interface of the Smart Hall Meal Management System

4.4.2 Student Dashboard

The Student Dashboard provides access to the features required for managing personal meal-related activities. Students can access meal booking, profile management, balance information, deposit requests, complaints, and notices.

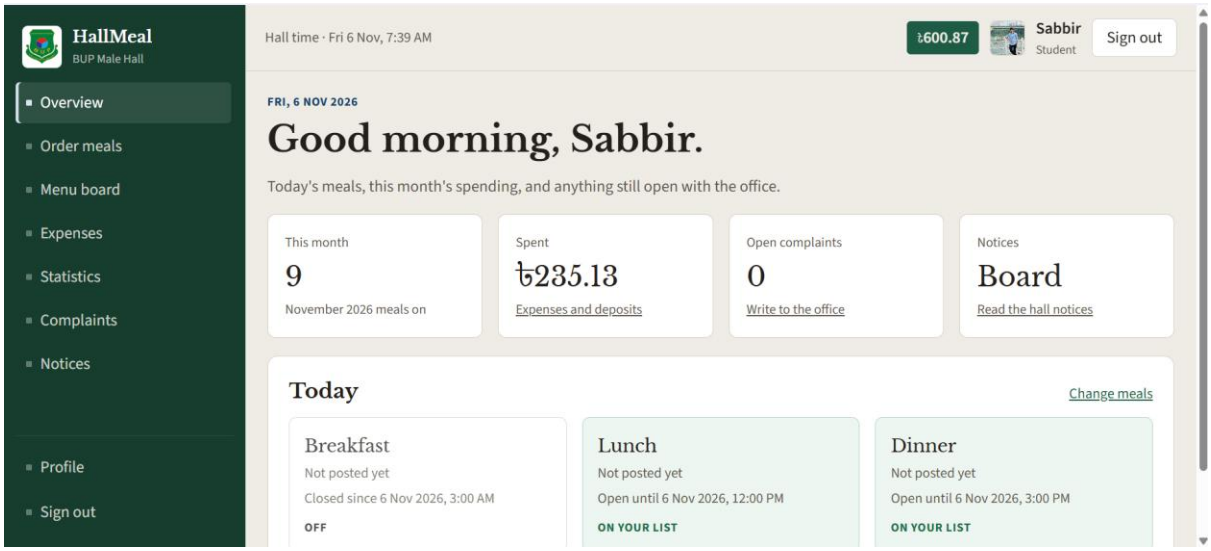


Figure 4.2: Student Dashboard

4.4.3 Admin Dashboard

The Admin Dashboard provides centralized access to the administrative features of the application. Admin can manage users, approve registrations, control meal settings, manage bazar expenses, review deposit requests, handle complaints, publish notices, and generate reports.

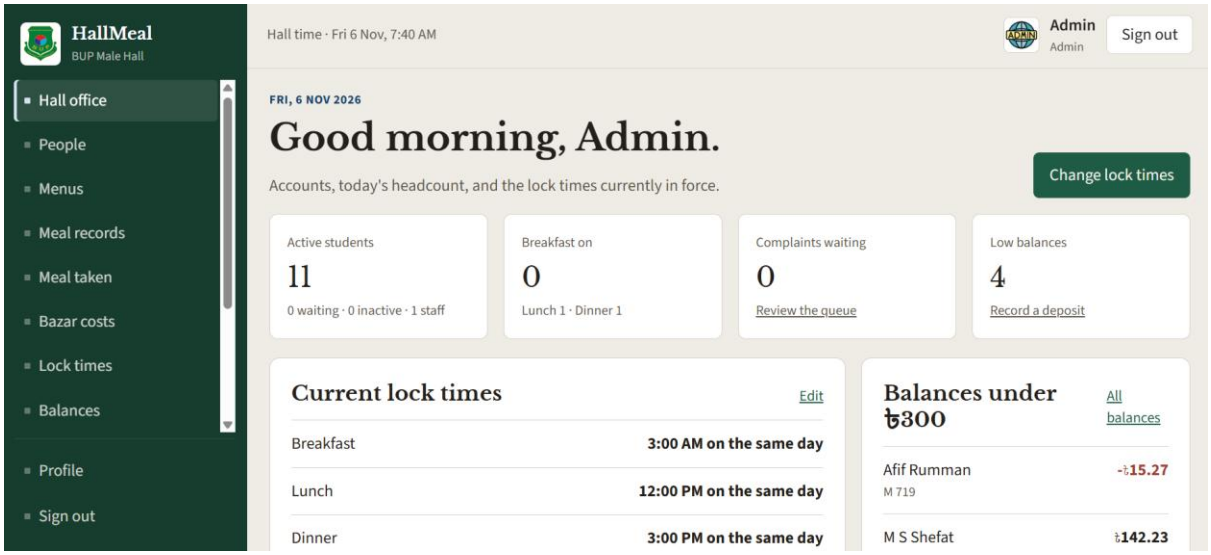


Figure 4.3: Admin Dashboard

4.4.4 Staff Dashboard

The Staff Dashboard provides access to the meal verification functionality. Staff members can use the QR code scanning feature to verify student meal eligibility and record meal-serving activities.

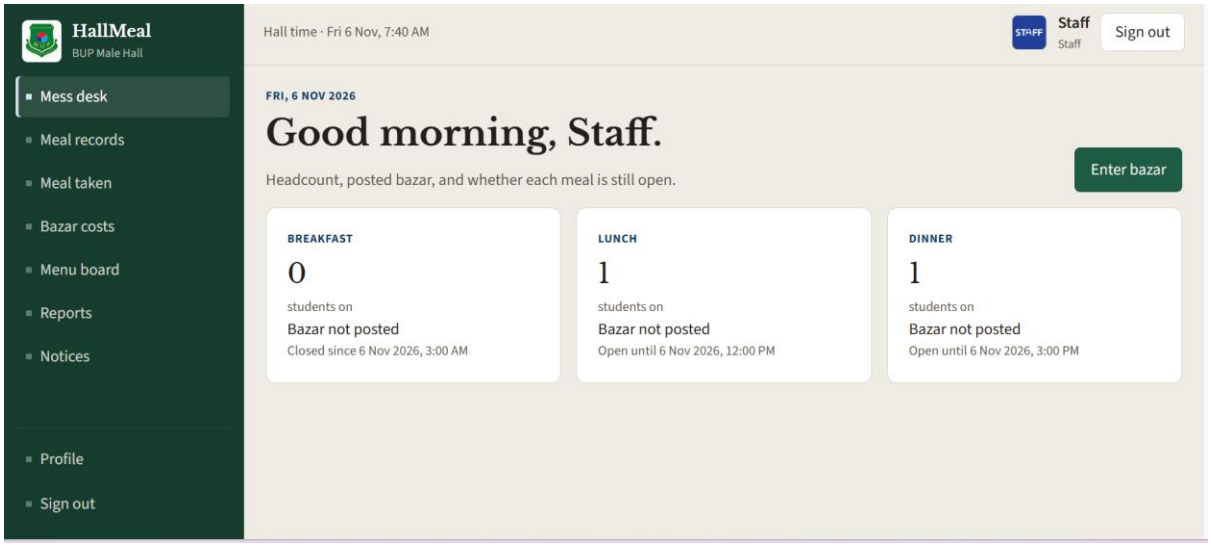


Figure 4.4: Staff Dashboard

4.5 Backend Implementation

The backend of the application was developed using Node.js and Express.js. It handles the application's business logic, processes client requests, communicates with the database, and manages authentication and authorization.

The backend performs several important operations, including:

1. Processing user registration and login requests.

2. Managing user sessions and authentication.
3. Verifying user roles and permissions.
4. Processing meal booking and cancellation requests.
5. Managing QR-based meal verification.
6. Recording daily bazar expenses.
7. Calculating meal rates.
8. Processing student balance and deposit requests.
9. Managing complaints and notices.
10. Generating reports and supporting CSV downloads.

The backend acts as the communication layer between the React.js frontend and the MongoDB database.

4.6 Database Implementation

MongoDB was selected as the database management system because of its flexible document-oriented data model.

The application uses MongoDB Atlas for cloud-based database hosting. The backend communicates with the database to store, retrieve, and update application information.

The database maintains information related to:

- User accounts and roles.
- Student meal bookings.
- Meal verification records.
- Daily bazar expenses.
- Meal rate information.
- Student balances and deposit requests.
- Complaints and administrative responses.
- Published notices.

The database integration ensures that information generated through different application modules can be stored and accessed when required.

4.7 Implementation of Major System Modules

This section describes the implementation of the major functional modules of the Smart Hall Meal Management System.

4.7.1 User Registration and Account Approval

The registration module allows students to create accounts by providing the required information. After registration, the account is submitted for administrative approval.

Admin can review the submitted registration information and approve or reject the request. This process helps ensure that only approved student accounts can access the relevant student features.

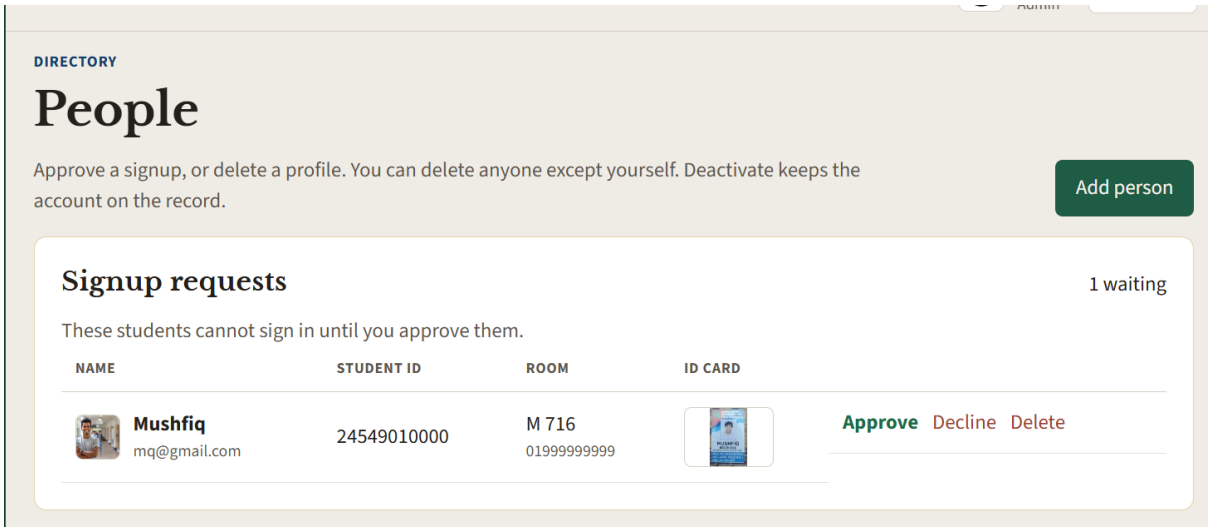


Figure 4.5: Student Registration and Account Approval Interface

4.7.2 Authentication and Session Management

The authentication module allows registered users to log in and access the application. The system uses session-based authentication to maintain the authenticated state of users.

After successful login, users receive access to the dashboard and functionalities associated with their assigned roles. The logout functionality allows users to terminate their active sessions.

The system also provides a temporary password reset mechanism for users who cannot access their accounts.

4.7.3 Meal Booking and Cutoff Management

The meal booking module allows students to select meals for available meal schedules. Students can also cancel previously selected meals within the permitted time.

Admin can configure the meal booking cutoff time. Once the deadline is reached, the system restricts further meal selection or cancellation according to the configured rules.

This module helps maintain organized meal records and provides an updated view of student meal selections.

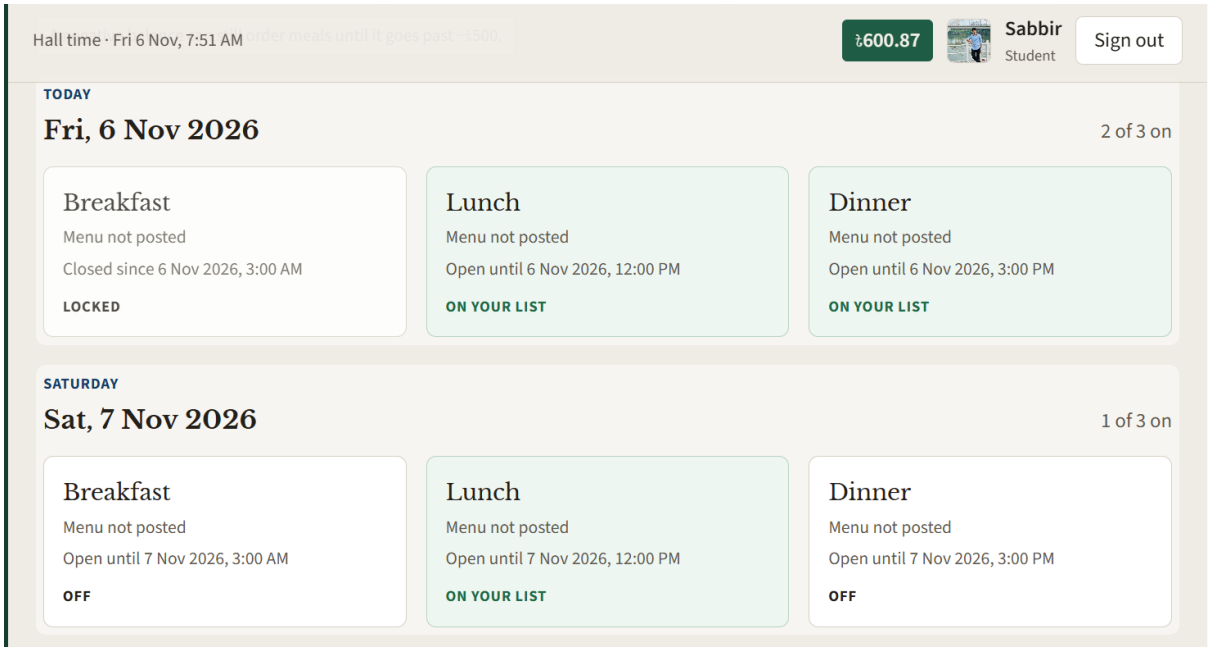


Figure 4.6: Student Meal Booking Interface

4.7.4 QR Code-Based Meal Verification

The QR code-based verification module is used during meal distribution. The system generates QR codes for the relevant meal verification process.

Staff members can scan student QR codes to verify whether the corresponding student has selected the meal. After successful verification, the system records the serving activity.

The verification process also helps prevent duplicate serving entries for the same meal record.

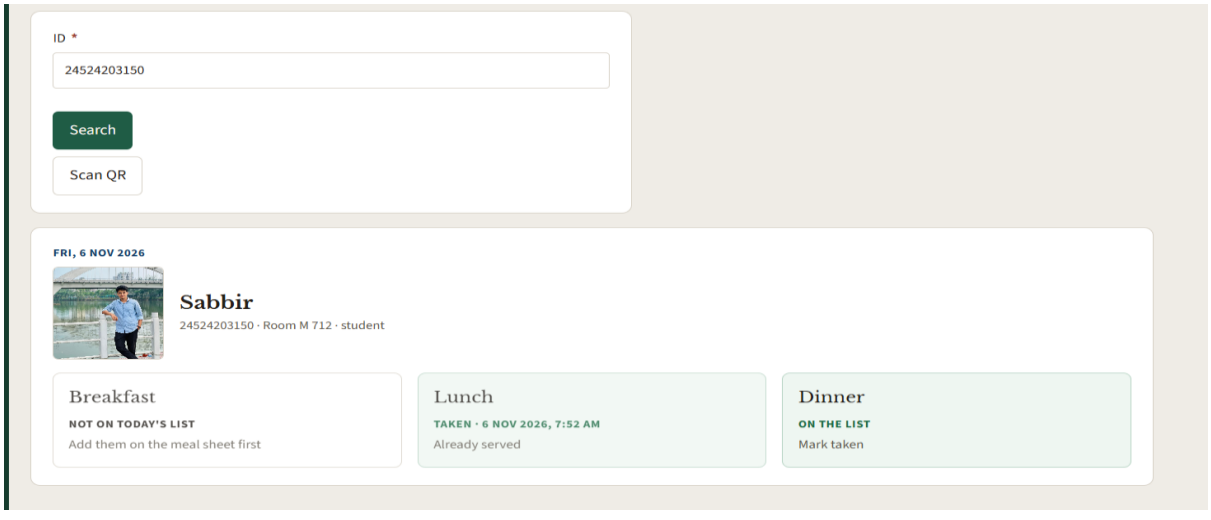


Figure 4.7: QR Code-Based Meal Verification Interface

4.7.5 Bazar Management and Meal Rate Calculation

The bazar management module allows authorized users to record daily bazar expenses. These records are used to calculate the meal rate.

The meal rate is calculated by dividing the relevant recorded bazar expense by the number of students who selected the corresponding meal.

The basic calculation is represented by the following formula:

Meal Rate = Total Bazar Expense / Total Selected Meals

If the division produces a fractional amount, the remaining amount is preserved in the calculation and reporting process.

This module helps maintain transparency in daily meal expenses and supports the calculation of individual meal costs.

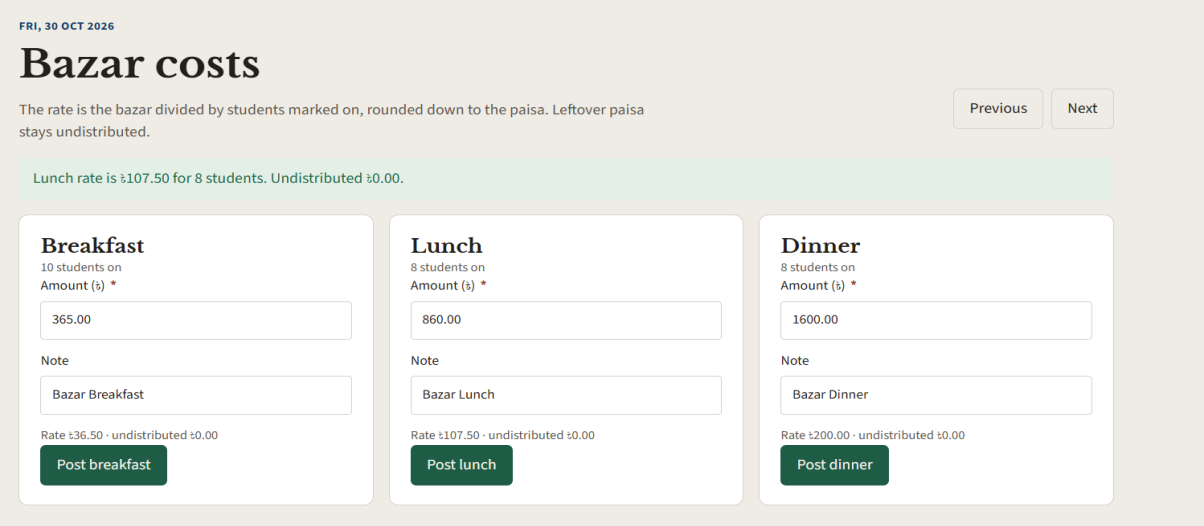


Figure 4.8: Bazar Expense Entry and Meal Rate Calculation Interface

4.7.6 Student Balance and Deposit Management

The balance management module allows students to view their account balance and monitor their financial information.

Students can submit deposit requests when they want to add money to their accounts. Admin can review these requests and approve or reject them.

After the administrative decision, the corresponding deposit information is reflected in the student's account records.

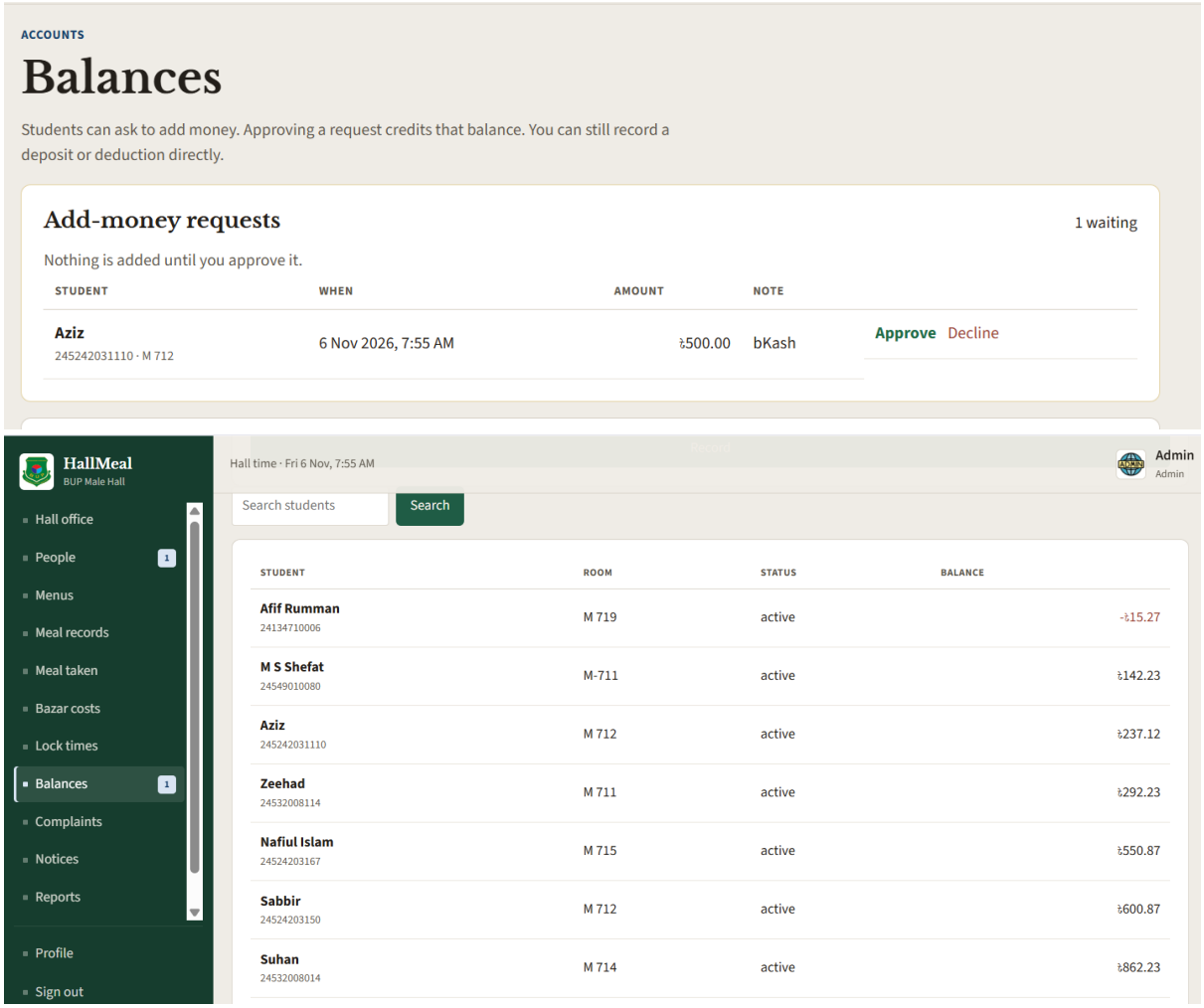


Figure 4.9: Balance and Deposit Management Interface

4.7.7 Complaint Management

The complaint management module provides a communication facility between students and the administration.

Students can submit complaints regarding hall meal-related issues. Admin can review the submitted complaints and provide responses through the application.

This module helps organize complaint records and provides a structured way to communicate concerns.

4.7.8 Notice Management

The notice management module allows Admin to publish important announcements through the application.

Students and other relevant users can view published notices from their respective interfaces. This feature provides a centralized platform for sharing important hall-related information.

4.7.9 Report Generation and CSV Export

The reporting module provides access to relevant meal, expense, and financial information.

Authorized users can generate reports based on the available application records. The system also supports downloading relevant reports in CSV format.

CSV export allows users to access the generated information in spreadsheet applications for further analysis and record-keeping.

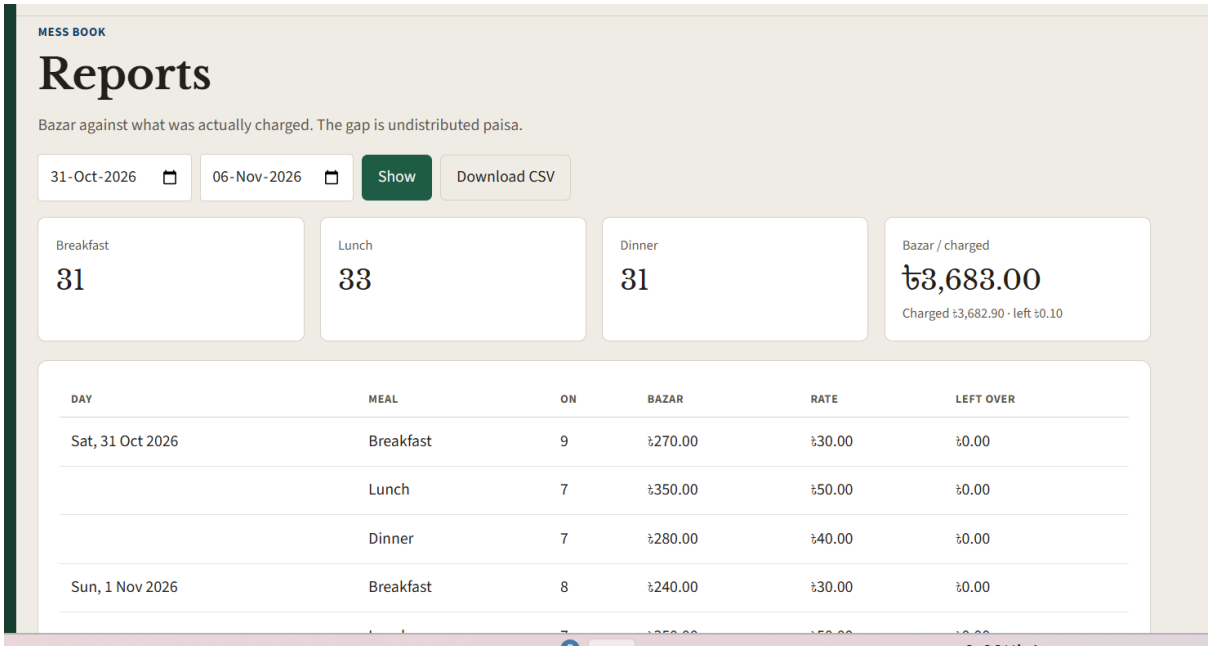


Figure 4.10: Report Generation and CSV Export Interface

4.8 Deployment and Hosting

After the implementation and testing of the application, the system was deployed to make it accessible through the internet.

The application is hosted on Render, while MongoDB Atlas is used for database hosting. The deployed application communicates with the cloud database to process user requests and maintain application records.

GitHub is used as the source code repository. The deployment environment is configured to support updates from the project repository through the deployment workflow.

The publicly accessible application can be visited through the following URL:

Project URL: https://smart-hall-meal.onrender.com

The deployment allows users to access the application using a modern web browser without requiring a local installation.

4.9 Summary

This chapter presented the implementation of the Smart Hall Meal Management System. It discussed the technologies and development environment used to build the application, including React.js, Node.js, Express.js, MongoDB, and MongoDB Atlas.

The chapter also described the implementation of the frontend, backend, database integration, authentication, meal booking, QR code-based verification, bazar management, meal rate calculation, balance management, complaint handling, notice management, and report generation.

Finally, the deployment process was discussed, explaining how the application was made accessible through Render. The next chapter focuses on the testing procedures, test results, and evaluation of the implemented system.

CHAPTER 5: SYSTEM TESTING AND RESULTS

5.1 Introduction

Software testing is an essential phase of software development. It is performed to identify errors, verify system functionality, and ensure that the application behaves according to its specified requirements.

The Smart Hall Meal Management System was tested to evaluate its major functionalities, user access permissions, input validation, and data management operations. Testing was conducted to verify whether the implemented features worked as expected under different user interactions.

This chapter presents the testing approach, functional testing results, validation and negative testing, role-based access control testing, database integrity testing, security testing, and overall evaluation of the system.

5.2 Testing Objectives

The primary objectives of testing the Smart Hall Meal Management System were:

1. To verify that all major system functionalities operate as expected.
2. To ensure that users can access only the features permitted by their assigned roles.
3. To validate user inputs and prevent invalid data submissions.
4. To verify the accuracy and consistency of stored information.
5. To evaluate the authentication and authorization mechanisms.
6. To identify potential errors and improve the overall reliability of the application.

5.3 Testing Methodology

The system was evaluated through manual testing. Different features were tested by interacting with the application through its user interface and observing the resulting system behavior.

The following testing approaches were considered:

Functional Testing: Verifies whether the implemented features perform their intended operations.

Validation Testing: Checks whether the system accepts valid information and handles invalid inputs appropriately.

Negative Testing: Evaluates how the system responds to incorrect, incomplete, or unauthorized operations.

Role-Based Access Control Testing: Checks whether Admin, Staff, and Student users can access only their permitted functionalities.

Database Integrity Testing: Checks whether important records remain accurate and consistent during system operations.

Security Testing: Evaluates the implemented authentication, authorization, and access restriction mechanisms.

5.4 Functional Testing

Functional testing was performed to verify the major features of the application. The test cases covered registration, authentication, meal booking, QR code verification, financial operations, complaint management, notice management, and reporting.

The following table summarizes the functional testing results.

Table 5.1: Functional Testing Results

Test ID	Test Scenario	Expected Result	Actual Result	Status
FT-01	Student Registration and Account Approval	Students can register, and Admin can approve or reject registration requests.	The registration and approval process worked as expected.	PASS
FT-02	Login, Logout and Forgot Password	Users can authenticate, terminate sessions, and use the available password reset process.	Authentication and password reset operations worked as expected.	PASS
FT-03	Meal Booking and Cutoff Time	Students can select or cancel meals within the permitted time.	Meal booking and cutoff restrictions worked as expected.	PASS
FT-04	QR Code Generation and Scanning	The system generates QR codes and allows Staff to scan them.	QR code generation and scanning worked as expected.	PASS
FT-05	Meal Serving Verification	The system verifies meal eligibility and records successful serving activities.	Meal verification and serving records worked as expected.	PASS

FT-06	Bazar Entry and Meal Rate Calculation	The system records bazar expenses and calculates meal rates.	Expense recording and meal rate calculation worked as expected.	PASS
FT-07	Balance, Deposit and Approval	Students can submit deposit requests, and Admin can approve or reject them.	Balance and deposit management operations worked as expected.	PASS
FT-08	Complaints and Admin Reply	Students can submit complaints, and Admin can provide responses.	Complaint submission and response operations worked as expected.	PASS
FT-09	Notice Management	Admin can publish notices, and relevant users can view them.	Notice management worked as expected.	PASS
FT-10	Reports and CSV Download	The system generates reports and supports CSV downloads.	Report generation and CSV download worked as expected.	PASS
FT-11	User Management and Permissions	Admin can manage user accounts and assigned permissions.	User management and permission-related operations worked as expected.	PASS
FT-12	Profile, Image and Password Update	Users can update permitted profile information, profile pictures, and passwords.	Profile and password update operations worked as expected.	PASS

The functional testing results indicate that all twelve major test scenarios passed during manual testing.

Table 5.2: Summary of Functional Testing

Testing Metric	**Result**
Total Test Scenarios	12
Passed Test Scenarios	12
Failed Test Scenarios	0
Pass Rate	100%

The pass rate was calculated using the following formula:

Pass Rate = (Passed Test Cases / Total Test Cases) × 100

Therefore,

Pass Rate = (12 / 12) × 100 = 100%

5.5 Validation and Negative Testing

Validation and negative testing were performed to examine how the application handles invalid inputs, incomplete information, and operations that should not be permitted.

The objective was to ensure that the system does not process invalid requests incorrectly and provides appropriate restrictions when users attempt unauthorized or invalid operations.

Table 5.3: Validation and Negative Testing

Test ID	Testing Scenario	Expected System Behaviour	Result
VT-01	Submitting incomplete registration information	The system should prevent submission of required incomplete information.	PASS
VT-02	Entering incorrect login credentials	The system should reject invalid authentication attempts.	PASS
VT-03	Accessing meal booking after the cutoff time	The system should restrict meal selection or cancellation after the configured deadline.	PASS
VT-04	Attempting duplicate meal verification	The system should prevent duplicate serving entries for the same meal record.	PASS
VT-05	Submitting invalid form information	The system should validate input data before processing the request.	PASS
VT-06	Attempting an unauthorized operation	The system should restrict operations that are not permitted for the user.	PASS

These tests helped evaluate the application's input validation and error-handling behaviour under different conditions.

5.6 Role-Based Access Control Testing

Role-Based Access Control (RBAC) testing was conducted to verify whether the system correctly restricts access according to the assigned user roles.

The system supports three primary roles: Admin, Staff, and Student. Each role is provided with specific permissions based on its responsibilities.

Table 5.4: Role-Based Access Control Testing

Test ID	Test Scenario	Expected Result	Status
RBAC-01	Admin accesses the administrative dashboard.	Admin can access permitted administrative functionalities.	PASS
RBAC-02	Staff accesses the meal verification interface.	Staff can access the permitted meal verification features.	PASS
RBAC-03	Student accesses the student dashboard.	Student can access permitted student functionalities.	PASS
RBAC-04	Student attempts to access administrative functions.	Unauthorized administrative access should be restricted.	PASS
RBAC-05	Staff attempts to access restricted administrative functions.	Access to restricted administrative operations should be denied.	PASS

The results indicate that the role-based access restrictions worked as expected during the performed tests.

5.7 Database Integrity Testing

Database integrity testing was performed to evaluate the consistency and correctness of information stored in the database during major application operations.

The testing focused on important data categories, including user accounts, meal bookings, meal verification records, bazar expenses, student balances, deposit requests, complaints, and notices.

Table 5.5: Database Integrity Testing

Test ID	Test Scenario	Expected Result	Status
DB-01	Saving new user information	Valid user information should be stored correctly.	PASS
DB-02	Updating user profile information	Updated information should be reflected in the corresponding user record.	PASS
DB-03	Recording meal bookings	Meal booking information should be stored accurately.	PASS
DB-04	Recording meal verification	Successful verification information should be stored correctly.	PASS

DB-05	Updating deposit information after approval	The corresponding balance information should be updated correctly.	PASS
DB-06	Recording bazar expenses	Expense information should be maintained correctly for meal rate calculation.	PASS

The performed tests indicated that the database operations associated with the selected application workflows behaved as expected.

5.8 Security Testing

Security testing was conducted to evaluate the application's authentication, authorization, and access restriction mechanisms.

The testing focused on ensuring that users must authenticate before accessing protected features and that system operations remain restricted according to assigned permissions.

The following areas were considered:

1. **Authentication:** Verified that users must provide valid credentials to access their accounts.
2. **Session Management:** Checked the login and logout processes and the handling of authenticated sessions.
3. **Authorization:** Evaluated whether different user roles can access only their permitted features.
4. **Unauthorized Access Prevention:** Checked whether users could access restricted application functions.
5. **Input Validation:** Evaluated the handling of invalid and incomplete user inputs.

The performed security checks helped assess the application's basic access protection mechanisms. However, these tests should not be considered a complete professional security audit or penetration test.

5.9 Overall Testing Results

The testing activities covered the major functional and operational areas of the Smart Hall Meal Management System.

The functional testing phase included twelve major test scenarios, all of which passed during manual testing. Additional testing activities were performed to evaluate validation, negative inputs, role-based access control, database integrity, and basic security mechanisms.

Table 5.6: Overall Testing Summary

Testing Category	Evaluation Result
Functional Testing	All 12 major test scenarios passed.
Validation Testing	Tested scenarios passed.
Negative Testing	Tested scenarios passed.
Role-Based Access Control Testing	Tested scenarios passed.
Database Integrity Testing	Tested scenarios passed.
Security Testing	Performed security checks passed.

The results demonstrate that the implemented features operated according to the expected behaviour during the performed tests.

5.10 Limitations of Testing

Although the system passed the performed functional and related testing activities, some areas may require further evaluation before large-scale deployment.

The following areas are identified for future testing and improvement:

1. **Load Testing:** Evaluating system performance under a large number of simultaneous users.
2. **Stress Testing:** Examining application behaviour under unusually high workloads.
3. **Cross-Browser Testing:** Performing broader compatibility testing across different browsers and devices for local host.
4. **Professional Security Assessment:** Conducting comprehensive vulnerability assessment and penetration testing.
5. **Backup and Recovery Testing:** Evaluating database backup procedures and recovery mechanisms.

These additional evaluations would help establish the application's readiness for broader operational use.

5.11 Summary

This chapter presented the testing procedures and results of the Smart Hall Meal Management System. Functional testing was performed on twelve major test scenarios, and all twelve scenarios passed during manual testing.

The chapter also discussed validation and negative testing, role-based access control testing, database integrity testing, and security testing. The results indicated that the tested features and access restrictions behaved as expected.

Finally, potential areas for further testing were identified, including load testing, stress testing, cross-browser compatibility, professional security assessment, and backup recovery evaluation.

The next chapter presents the conclusion of the project, its major contributions, limitations, and possible future improvements.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Introduction

The Smart Hall Meal Management System was developed to simplify and organize the daily meal management activities of a university hall. The project focuses on reducing manual operations, improving record management, and providing a centralized platform for students, staff, and administrators.

The system integrates several essential functionalities, including student registration, meal booking, QR code-based meal verification, bazar expense management, meal rate calculation, balance management, complaint handling, notice publication, and report generation.

This chapter presents the overall conclusion of the project, its major achievements, existing limitations, and possible future improvements.

6.2 Project Achievements

The development of the Smart Hall Meal Management System resulted in a web-based application that provides an organized approach to managing hall meal-related activities.

The major achievements of the project are described below:

6.2.1 Centralized Meal Management

The system provides a centralized platform for managing meal-related information. It allows students and administrators to access relevant information through a single application.

6.2.2 Role-Based User Management

The application supports three primary user roles: Admin, Staff, and Student. Each role has specific permissions and functionalities, helping maintain organized access to different system operations.

6.2.3 Online Meal Booking

Students can select or cancel meals through the application within the configured cutoff time. This reduces the need for manual meal selection records.

6.2.4 QR Code-Based Meal Verification

The QR code-based verification feature allows Staff to verify student meal eligibility during meal distribution. It also helps prevent duplicate serving entries.

6.2.5 Automated Meal Rate Calculation

The system records bazar expenses and calculates meal rates based on the number of students who selected the corresponding meal. This helps simplify expense calculation and financial record management.

6.2.6 Digital Balance and Deposit Management

Students can view their account balances and submit deposit requests through the application. Admin can review and approve or reject these requests, helping maintain organized financial records.

6.2.7 Complaint and Notice Management

The application provides a structured communication process through complaint submission, administrative responses, and notice publication.

6.2.8 Report Generation

The reporting module allows authorized users to generate relevant reports and download supported information in CSV format. This helps with record-keeping and further data analysis.

6.3 Project Evaluation

The implemented system was evaluated through manual functional testing and additional testing activities.

Twelve major functional test scenarios were performed, and all twelve passed. Validation testing, negative testing, role-based access control testing, database integrity testing, and basic security checks were also conducted.

The testing results indicate that the implemented features operated according to their expected behaviour during the performed tests.

The project demonstrates how modern web technologies can be used to develop a centralized platform for managing university hall meal activities.

6.4 Limitations of the System

Although the Smart Hall Meal Management System provides several useful functionalities, some limitations remain that may be addressed in future versions.

The major limitations are listed below:

1. **No Integrated Online Payment Gateway:** The current version does not include a direct online payment gateway. Deposit requests are managed through the application's administrative approval process.
2. **Limited Performance Evaluation:** Extensive load testing and stress testing have not been conducted to evaluate system behaviour under a large number of simultaneous users.
3. **Limited Security Assessment:** Basic security checks have been performed, but a comprehensive professional security audit and penetration testing are outside the scope of the current project.
4. **Limited External Integration:** The current application does not include extensive integration with external university management systems or payment services.
5. **Dependence on Internet Connectivity:** Since the application is web-based and deployed online, users require an internet connection to access its features.

These limitations provide opportunities for further development and improvement.

6.5 Future Work

Several improvements can be introduced in future versions to make the Smart Hall Meal Management System more comprehensive, efficient, and suitable for broader use.

6.5.1 Online Payment Integration

An online payment gateway can be integrated to allow students to deposit money directly through supported digital payment services. This would reduce the need for manual deposit processing.

6.5.2 Mobile Application Development

A dedicated Android and iOS application can be developed to provide easier access to meal booking, balance information, notices, and other features through mobile devices.

6.5.3 Advanced Analytics and Visualization

Advanced analytical features can be introduced to generate visual reports and statistical summaries of meal consumption, daily expenses, student balances, and monthly financial activities.

6.5.4 Automated Notifications

Email, SMS, or push notification services can be integrated to notify students about important announcements, deposit approvals, meal booking deadlines, and other relevant updates.

6.5.5 Enhanced Security Mechanisms

Additional security mechanisms, such as login rate limiting, stronger session configuration, multi-factor authentication, and regular security assessments, can be implemented to improve application security.

6.5.6 Advanced Reporting System

The reporting module can be expanded to support more detailed financial summaries.

6.5.7 Backup and Recovery Mechanism

An automated database backup and recovery mechanism can be introduced to improve data protection and reduce the risk of information loss.

6.5.8 Integration with University Systems

In the future, the application can be integrated with existing university or hall management systems to improve student information management and administrative coordination.

6.6 Conclusion

The Smart Hall Meal Management System was developed to address common challenges associated with traditional hall meal management. The application provides a centralized platform that combines meal booking, QR code-based verification, expense recording, meal rate calculation, balance management, deposit approval, complaint handling, notice publication, and report generation.

The use of React.js, Node.js, Express.js, and MongoDB enabled the development of a modern web-based application with separate interfaces for Admin, Staff, and Student users.

The performed testing activities demonstrated that the major implemented functionalities operated according to their expected behaviour. The system therefore provides a practical foundation for improving the organization and management of hall meal-related activities.

Although some limitations remain, the project establishes opportunities for future improvements through online payment integration, mobile application development, advanced reporting, enhanced security, and external system integration.

Overall, the Smart Hall Meal Management System demonstrates the practical application of web technologies and software engineering concepts to address a real-world management problem in a university hall environment.

REFERENCES

1. React.js Official Documentation. https://react.dev/learn
2. Node.js Official Documentation. https://nodejs.org/docs/latest/api/
3. Express.js Official Documentation. https://expressjs.com/
4. MongoDB Official Documentation. https://www.mongodb.com/docs/
5. MongoDB Atlas Documentation. https://www.mongodb.com/docs/atlas/
6. Render Official Documentation. https://render.com/docs
7. GitHub Documentation. https://docs.github.com/
8. Git Official Documentation. https://git-scm.com/doc
9. Smart Hall Meal Management System – Project Repository. https://github.com/msshefat/Smart-Hall-Meal-Management-System
10. Smart Hall Meal Management System – Live Application. https://smart-hall-meal.onrender.com